

Worksheet-2



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Tasks

Task 1: Basic student grading system prototype using classes and objects.

Write a program that manages a simple student grade calculator with the following requirements.

Create a `Student` class that has:

1. Student name (string)
2. Three subject marks (integers)
3. A basic member function to calculate average

```
#include <iostream>
```

```
using namespace std;
```

```
class Student {
```

```
public:
```

```
    string name;
```

```
    int mark1, mark2, mark3;
```

```
    float total;
```

```
public:
```

```
    void getinput()
```

```
{
```

```
    cout<<"Enter student name"<<endl;
```

```
    cin>>name;
```

```
    cout<<"Enter marks for subject 1"<<endl;
```

```
    cin>>mark1;
```

```
    cout<<"Enter marks for subject 2"<<endl;
```

```
    cin>>mark2;

    cout<<"Enter marks for subject 3"<<endl;

    cin>>mark3;

}

float getTotal()

{

    total= mark1 + mark2 + mark3;

}

float getPercentage()

{

    return getTotal() / 3.0;

}

void check()

{

    if(mark1<0 || mark1>100 || mark2<0 || mark2>100 || mark3<0 || mark3>100 )

    {

        cout<<"Invalid input of marks."<<endl;

        void getinput();

    }

}
```

```
}
```

```
float getGPA() {  
  
    float avg = getPercentage();  
  
    if (avg >= 90) return 4.0;  
  
    else if (avg >= 80) return 3.0;  
  
    else if (avg >= 70) return 2.0;  
  
    else if (avg >= 60) return 1.0;  
  
    else return 0.0;  
  
}
```

```
char getGrade() {  
  
    float avg = getPercentage();  
  
    if (avg >= 90) return 'A';  
  
    else if (avg >= 80) return 'B';  
  
    else if (avg >= 70) return 'C';  
  
    else if (avg >= 60) return 'D';  
  
    else return 'F';  
  
}
```

```
bool isValidMarks() {  
  
    return (mark1 >= 0 && mark1 <= 100) &&  
  
        (mark2 >= 0 && mark2 <= 100) &&
```

```
(mark3 >= 0 && mark3 <= 100);

}

};

int main() {

    Student s;

    s.getinput();

    s.check();

    float getTotal=s.getTotal();

    float percentage = s.getPercentage();

    float gpa = s.getGPA();

    char grade = s.getGrade();


    cout << "\nStudent Name: " << s.name << endl;

    cout << "Total Marks: " << s.total << endl;

    cout << "Percentage: " << percentage << "%" << endl;

    cout << "GPA (4.0 scale): " << gpa << endl;

    cout << "Grade: " << grade << endl;


    return 0;

}
```

Output:

```
"C:\Users\Aashr\Desktop\Co... x + v
Enter student name
Aashruti
Enter marks for subject 1
89
Enter marks for subject 2
98
Enter marks for subject 3
78

Student Name: Aashruti
Total Marks: 265
Percentage: 88.3333%
GPA (4.0 scale): 3
Grade: B

Process returned 0 (0x0)   execution time : 20.724 s
Press any key to continue.
|
```

Task 2: Programming assignments: All questions are mandatory

1. Write a program with a class `Circle` having:
 1. Private member: `radius` (float)
 2. A constructor to initialize `radius`
 3. A friend function `compareTwoCircles` that takes two `Circle` objects and prints which circle has the larger area

```
#include<iostream>
```

```
using namespace std;
```

```
class Circle {
```

```
private:
```

```
    float radius;
```

```
public:
```

```
    Circle(float r) {
```

```
        radius = r;
```

```
    }
```

```
    float getArea() const {
```

```
        return 3.14159 * radius * radius;
```

```
    }
```

```
    friend void compareTwoCircles(const Circle& c1, const Circle& c2);
```

```
};
```

```
void compareTwoCircles(const Circle& c1, const Circle& c2) {
```

```
    float area1 = c1.getArea();
```

```
    float area2 = c2.getArea();
```

```
    if (area1 > area2) {  
        cout << "First circle has a larger area." << endl;  
    } else if (area2 > area1) {  
        cout << "Second circle has a larger area." << endl;  
    } else {  
        cout << "Both circles have equal area." << endl;  
    }  
}  
  
int main() {  
    float radius1, radius2;  
  
    cout << "Enter radius of first circle: ";  
  
    cin >> radius1;  
  
    cout << "Enter radius of second circle: ";  
  
    cin >> radius2;  
  
    Circle circle1(radius1);  
  
    Circle circle2(radius2);  
  
    compareTwoCircles(circle1, circle2);  
  
    return 0;  
}
```


Output

```
"C:\Users\aaashr\Desktop\Co... X + v
Enter radius of first circle: 2.3
Enter radius of second circle: 4.5
Second circle has a larger area.

Process returned 0 (0x0)   execution time : 5.491 s
Press any key to continue.
|
```

2. Create a program with these overloaded functions named `findMax`:
1. One that finds maximum between two integers
 2. One that finds maximum between two floating-point numbers
 3. One that finds maximum among three integers
 4. One that finds maximum between an integer and a float

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
class error {
```

```
public:
```

```
    string what() {
```

```
        return "Invalid input detected!";
```

```
    }
```

```
};
```

```
class Students {
```

```
protected:
```

```
    int Rollno;
```

```
    string Name;
```

```
    float Marks;
```

```
public:
```

```
    void getinput() {
```

```
cout << "Enter Student Details: " << endl;
```

```
cout << "Student Roll No: ";
```

```
cin >> Rollno;
```

```
cout << "Student Name: ";
```

```
cin >> Name;
```

```
cout << "Marks obtained: ";
```

```
cin >> Marks;
```

```
if (Marks < 0 || Marks > 100) {
```

```
    throw error();
```

```
}
```

```
}
```

```
void display() {
```

```
    cout << "Roll No: " << Rollno << endl;
```

```
    cout << "Name: " << Name << endl;
```

```
    cout << "Marks: " << Marks << endl;
```

```
    cout << "======" << endl;
```

```
}
```

```
void readfile() {
```

```
    ifstream outFile("Student.txt");
```

```
while (outFile >> Rollno >> Name >> Marks) {  
  
    if (Marks < 0 || Marks > 100) {  
  
        throw string("Invalid marks found in file!");  
  
    }  
  
    display();  
  
}
```

```
outFile.close();  
  
}
```

```
void writefile() {  
  
    ofstream outFile("Student.txt", ios::app);  
  
    outFile << Rollno << " " << Name << " " << Marks << endl;  
  
    outFile.close();  
  
}  
  
};
```

```
int main() {  
  
    try {  
  
        Students s1;  
  
        s1.getinput();
```

```
s1.writefile();

s1.readfile();

cout << "New record added successfully." << endl;

}

catch (error &e) {

    cout << e.what() << endl;

}

catch (string &msg) {

    cout << msg << endl;

}

return 0;

}
```

Output

```
"C:\Users\aaashr\Desktop\Coli x + v
Enter Student Details:
Student Roll No: 2
Student Name: bina
Marks obtained: 99
Roll No: 1
Name: Aashruti
Marks: 98
=====
Roll No: 2
Name: bina
Marks: 99
=====
New record added successfully.

Process returned 0 (0x0)   execution time : 12.168 s
Press any key to continue.
|
```

Task 3: Basics of File Handling

Write a program that reads the titles of 10 books (use an array of 150 characters) and writes them in a binary file selected by the user. The program should read a title and display a message to indicate if it is contained in the file or not.

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <cstring>
```

```
using namespace std;
```

```
class BookManager
```

```
{
```

```
    char titles[10][150];
```

```
    char filename[100];
```

```
public:
```

```
    void getFilename()
```

```
    {
```

```
        cout << "Enter the binary file name: "<<endl;
```

```
        cin >> filename;
```

```
    }
```

```
    void inputTitles()
```

```
    {
```

```
        cout << "Enter 10 book titles:"<<endl;
```

```
        for (int i = 0; i < 10; i++)
```

```
        {
```

```
        cout << "Title " << (i + 1) << ": ";

        cin >> titles[i];

    }

}

void writeToFile()

{

    ofstream out(filename, ios::binary);

    if (!out)

    {

        cout << "Error!!";

        return;

    }

    for (int i = 0; i < 10; i++)

    {

        out.write(titles[i], sizeof(titles[i]));

    }

    out.close();

    cout << "Book titles saved to file."<<endl;

}

void searchTitle()

{
```



```
char searchbook[150];  
  
cout << "Enter the book title to search: "<<endl;  
  
cin >> searchbook;
```

```
ifstream in(filename, ios::binary);  
  
if (!in) {  
    cout << "Error!!"<<endl;  
    return;  
}
```

```
char temp[150];  
  
bool found = false;
```

```
while (in.read(temp, sizeof(temp))) {  
    if (strcmp(temp, search) == 0) {  
        found = true;  
        break;  
    }  
}
```

```
in.close();
```

```
if (found)  
    cout << "Book found in the file."  
  
else
```

```
        cout << "Book not found in the file.";
    }
};
```

```
int main() {
    BookManager b;
    b.getFilename();
    b.inputTitles();
    b.writeToFile();
    b.searchTitle();

    return 0;
}
```

Output:

```
"C:\Users\ashr\Desktop\Co" × + v
Enter the binary file name:
1
Enter 10 book titles:
Title 1: life
Title 2: motive
Title 3: health
Title 4: mindset
Title 5: spare
Title 6: asus
Title 7: difference
Title 8: oneyear
Title 9: split
Title 10: wood
Book titles saved to file.
Enter the book title to search:
life
Book found in the file.
Process returned 0 (0x0)   execution time : 96.139 s
Press any key to continue.
|
```

2. Create a program that:

1. Reads student records (roll, name, marks) from a text file
2. Throws an exception if marks are not between 0 and 100
3. Allows adding new records with proper validation
4. Saves modified records back to file
5. Saves modified records back to file

```
#include <iostream>
#include <fstream>
using namespace std;

class error {
public:
    string what() {
        return "Invalid input detected!";
    }
};

class Students {
protected:
    int Rollno;
    string Name;
    float Marks;

public:
    void getinput() {
        cout << "Enter Student Details: " << endl;
        cout << "Student Roll No: ";
        cin >> Rollno;
        cout << "Student Name: ";
        cin >> Name;
        cout << "Marks obtained: ";
        cin >> Marks;

        if (Marks < 0 || Marks > 100) {
            throw error();
        }
    }

    void display() {
        cout << "Roll No: " << Rollno << endl;
        cout << "Name: " << Name << endl;
    }
};
```

```
        cout << "Marks: " << Marks << endl;
        cout << "======" << endl;
    }

    void readfile() {
        ifstream outFile("Student.txt");

        while (outFile >> Rollno >> Name >> Marks) {
            if (Marks < 0 || Marks > 100) {
                throw string("Invalid marks found in file!");
            }
            display();
        }

        outFile.close();
    }

    void writefile() {
        ofstream outFile("Student.txt", ios::app);
        outFile << Rollno << " " << Name << " " << Marks << endl;
        outFile.close();
    }
};

int main() {
    try {
        Students s1;
        s1.getinput();
        s1.writefile();
        s1.readfile();
        cout << "New record added successfully." << endl;
    }
    catch (error &e) {
        cout << e.what() << endl;
    }
    catch (string &msg) {
        cout << msg << endl;
    }

    return 0;
}
```

Output:

```
"C:\Users\aaashr\Desktop\Co  X  +  v
Enter Student Details:
Student Roll No: 1
Student Name: Aashruti
Marks obtained: 98
Roll No: 1
Name: Aashruti
Marks: 98
=====
Roll No: 2
Name: bina
Marks: 99
=====
Roll No: 1
Name: Aashruti
Marks: 98
=====
New record added successfully.

Process returned 0 (0x0)   execution time : 38.558 s
Press any key to continue.
|
```

	File	Edit	View
1	<u>Aashruti</u>	98	
2	<u>bina</u>	99	
1	<u>Aashruti</u>	98	

